# Alvaro Pastor, PhD

XR researcher and developer | Episodic memory, immersive systems, generative AI
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## 1. EDUCATION

**2026 Ph.D., Network and Information Technologies**
Open University of Catalonia, Spain.
*Dissertation: "Virtual and Augmented Reality Investigations of Episodic Memory: The Role of Locomotion, Spatial Features and Emotion".*

**2015 M.Sc., Cognitive Systems and Interactive Media**
Pompeu Fabra University, Spain.

**2013 Architect**
Applied Sciences University, Peru.

## 2. PROFESSIONAL APPOINTMENTS

**2026–present. Technical Specialist, Extended Reality Design & Development**
XRLab, Interdisciplinary Research and Innovation Center, Open University of Catalonia, Spain

**2020–2026. Professor, Master's Degree in Videogame Design & Development**
Department of Computer Science, Multimedia, and Telecommunication, Open University of Catalonia, Spain

**2023–2024. Technical Specialist, Virtual Reality Design & Development**
EventLAB, Department of Clinical Psychology and Psychobiology, University of Barcelona, Spain.

**2022–2024. Professor, Degree in Videogame Design & Development**
Centre for Image and Multimedia Technology, Polytechnic University of Catalonia, Spain.

## 3. GRANTS & FUNDED RESEARCH

**2024–2027 Inhabiting the Hybrid**: **Contributions from Artistic Research in Interactive and Immersive Media.** Ministry of Science and Innovation & European Regional Development Fund (PID2021-128875NA-I00). *Role: Researcher. Status: Active.*

**2024–2025 NeuroScent VR.** Research Accelerator Grant, Open University of Catalonia (RA405). *Role: Lead Designer and Developer. Status: Completed.*

**2023–2024 EnvironMENTAL.** Horizon Europe (101057429). University of Barcelona. *Role: Lead Virtual Reality Design & Development. Status: Completed.*

**2021–2022 The Memory Palace.** Innovative Information and Communication Technologies Grant Barcelona Culture Institute (21S03773-006). *Role: Principal Investigator. Status: Completed.*

**2019 Industrial Doctorates Grant.** Catalan Agency for Management of University and Research Grants. *Role: PhD Fellow. Status: Awarded, declined.*

## 4. PEER-REVIEWED PUBLICATIONS

**Pastor, A.; Bourdin-Kreitz, P. (2026). Designing memorable XR under stress: An empirically grounded criterion for long-term memory binding.** *ACM Conference on Human Factors in Computing Systems (CHI 2026). Workshop on XR for Challenging Environments: Enabling Human Performance and Agency under Stress (XR4CE), April 14, 2026, Barcelona, Spain. CEUR Workshop Proceedings, Vol-4256, pp. 1–5. URL: https://ceur-ws.org/Vol-4256/XR4CE_paper_2.pdf*

**Pastor, A.; Bourdin-Kreitz, P. (2026). Comparing Episodic Memory Binding Outcomes After Emotion Induction in Virtual Reality.** *Virtual Reality*. https://doi.org/10.1007/s10055-026-01364-9

**Pastor, A.; Bourdin-Kreitz, P. (2026, under review). Synthetic generation of photorealistic 3D characters for face-based episodic memory measures.** *IEEE Open Journal of the Computer Society.*

**Pastor, A.; Bourdin-Kreitz, P. (2025). Synthetic generation of photorealistic 3D characters for face-based episodic memory measures.** Preprint, *OSF*. https://doi.org/10.31219/osf.io/4jf2g_v1

**Pastor, A. (2025). Augmenting Reality: On the Shared History of Perceptual Illusion and Video Projection Mapping.** *Estudis Escènics - Quaderns del Institut del Teatre*, 50, 326 - 343. e-ISSN: 2385-362X.

**Pastor, A.; Bourdin-Kreitz, P. (2024). Comparing episodic memory outcomes from walking augmented reality and stationary virtual reality encoding experiences.** *Scientific Reports, 14*, 1-23. https://doi.org/10.1038/s41598-024-57668-w

## 5. TECHNICAL & METHODOLOGICAL EXPERTISE

**Computational Neuroscience:** fMRI preprocessing (fMRIPrep, FSL) and statistical analysis (Python, Nilearn) using open neuroimaging datasets.

**Machine Learning & Computer Vision:** Deep learning frameworks (PyTorch, TensorFlow) for generative photorealistic face synthesis, validated in published memory research. Generative pipelines for experimental stimulus creation.

**Immersive & Interactive Systems:** Augmented and Virtual Reality experimental design and development, spanning locomotion-based and stationary paradigms. Interactive systems architecture for precision-timed multimodal stimulus delivery.

**Cognitive Science Methods:** Design and psychometric validation of ecologically valid cognitive assessments for immersive environments. Integration of standard psychometric frameworks.

**Hardware Integration:** Psychophysiological sensor integration and multimodal data synchronisation. Electromechanical systems engineering for experimental platforms. High-performance computing infrastructure assembly.

**Data Science:** Statistical analysis and computational modeling of behavioural and physiological data (Python, JASP/SPSS). Scientific data visualisation.